Leg Pain on Walking

CLINICAL CHALLENGE 01 | EDITORIAL REVIEW COPY

Current wording, exported 18 October 2026. Reference IDs identify passages for requested edits. Burgundy bands show inquiry buttons; the response appears beneath. Grey notes explain display conditions. Wording is reproduced without editorial changes.

P01 | Patient presentation / initial referral

A 54-year-old man reports gradually worsening pain in both legs over the past year. He has begun cutting back on walks and errands.

01 / History inquiries - first four

H01 | History

What triggers the pain?

Stored full question: What brings on the pain?

Walking or standing for several minutes brings on aching and heaviness in his legs.

H02 | History

What relieves it?

Stored full question: What makes it better?

Sitting usually relieves symptoms within a few minutes. Simply stopping while standing helps much less.

H03 | History

Does leaning forward help?

Yes. Leaning over a shopping cart lets him walk much farther.

H04 | History

Where does the pain spread?

Stored full question: Where does the discomfort travel?

It involves both buttocks and the back of the thighs, sometimes extending into the calves.

EDITORIAL NOTES / REPLACEMENT WORDING

02 / History inquiries - remaining four

H05 | History

Does slope affect walking?

Stored full question: Does walking uphill or downhill differ?

Downhill walking is more difficult than uphill walking.

H06 | History

Can he cycle?

Stored full question: Can he ride a bicycle?

He tolerates a stationary bicycle much better than upright walking.

H07 | History

Any vascular risk factors?

He has hypertension and previously smoked; he has no known peripheral arterial disease.

H08 | History

Any neurological red flags?

Stored full question: Any new bladder, saddle, or progressive weakness symptoms?

No bowel or bladder changes, saddle sensory changes, or progressive focal weakness.

EDITORIAL NOTES / REPLACEMENT WORDING

03 / Examination and imaging inquiries

E01 | Examination

How are the leg pulses?

Both feet are warm, and distal pulses are palpable bilaterally.

E02 | Examination

Neurological exam findings?

Stored full question: What is the neurological examination at rest?

Lower-extremity strength, reflexes, and light-touch sensation are broadly normal at rest.

E03 | Examination

Straight leg raise result?

Stored full question: What does straight leg raise show?

Straight leg raise does not reproduce his characteristic symptoms.

E04 | Examination

Lumbar movement findings?

Stored full question: Does lumbar movement provoke symptoms?

Extension is restricted and increases discomfort; flexion is better tolerated.

I01 | Imaging

Previous lumbar imaging?

Stored full question: Any previous lumbar MRI or CT?

He has not had a lumbar MRI or CT.

EDITORIAL NOTES / REPLACEMENT WORDING

04 / Clinical debrief - diagnosis and learning points

Shown after a correct diagnosis or two unsuccessful attempts. Learner-entered diagnoses are reproduced from their input rather than authored case text.

D01 | Debrief title

Diagnosis & Debrief

Shown after a correct diagnosis.

Clinical Debrief

Shown after two unsuccessful attempts.

D02 | Working diagnosis and explanation

Best working diagnosis: Lumbar spinal stenosis causing neurogenic claudication

This pattern is most consistent with neurogenic claudication due to lumbar spinal stenosis. The key is posture-dependent leg discomfort brought on by standing or walking and relieved by sitting or lumbar flexion. The clinical pattern is suggestive rather than imaging confirmation.

D03 | Learner response headings

Your final diagnosis

Followed by the learner's final submitted text.

Your ranked differential

Shown only if entries were provided; followed by their ranked list.

D04 | Key learning points

Flexion-related relief and the shopping-cart pattern help distinguish neurogenic from vascular claudication.

Normal resting neurological findings do not exclude lumbar spinal stenosis.

Assess vascular risk and red flags rather than relying on one characteristic clue.

D05 | Clinical caveat

A clinical working diagnosis; imaging is considered in context, especially if symptoms are persistent or intervention is contemplated.

EDITORIAL NOTES / REPLACEMENT WORDING

05 / Clinical debrief - findings and missed inquiries

D06 | Diagnostically useful findings

D06-H01 | Standing and walking provocation

Standing and walking provocation: Walking or standing for several minutes brings on aching and heaviness in his legs. The activity pattern matters, but exertional pain alone does not distinguish a neurological from a vascular cause.

D06-H02 | Sitting versus standing rest

Sitting versus standing rest: Sitting usually relieves symptoms within a few minutes. Simply stopping while standing helps much less. The difference between sitting and simply stopping upright helps establish whether posture changes the symptoms.

D06-H03 | Relief with forward flexion

Relief with forward flexion: Yes. Leaning over a shopping cart lets him walk much farther. The shopping-cart response adds a posture-dependent clue; it should be interpreted with the rest of the assessment.

D06-E01 | Vascular examination

Vascular examination: Both feet are warm, and distal pulses are palpable bilaterally. Pulses contribute to the vascular assessment but should not be used alone to exclude peripheral arterial disease.

D06-H08 | Neurological red-flag screen

Neurological red-flag screen: No bowel or bladder changes, saddle sensory changes, or progressive focal weakness. New bowel or bladder dysfunction, saddle symptoms, or progressive weakness would change the urgency of assessment.

D07 | Potentially useful findings not investigated

For each unasked finding from D06, the game displays that finding's title and explanation (the "why" sentence), without repeating its clinical response. Only unasked findings are listed.

If all five findings have been investigated, the game displays:

You reviewed the main discriminating findings highlighted in this case. Further inquiry should address a specific unresolved question.

EDITORIAL NOTES / REPLACEMENT WORDING

06 / Clinical debrief - alternatives and next steps

D08 | Important alternatives

D08-1 | Peripheral arterial disease / vascular claudication

Peripheral arterial disease / vascular claudication: Compare relief from standing rest with relief from sitting or flexion. Assess vascular risk and the leg examination; ankle-brachial index testing may be appropriate when PAD is suspected.

D08-2 | Bilateral lumbar radicular pain

Bilateral lumbar radicular pain: Check for a consistent nerve-root distribution, neural tension signs, and focal neurological deficits rather than inferring a root lesion from leg pain alone.

D08-3 | Peripheral neuropathy

Peripheral neuropathy: Look for persistent distal sensory changes and a peripheral distribution rather than using walking intolerance alone to decide.

D09 | Next clinical steps

D09-1 | Next step

Assess the effect on walking and daily function, complete the neurological and vascular examination, and reassess any red flags.

D09-2 | Next step

Consider imaging in clinical context, particularly if symptoms persist or intervention is contemplated. There is no MRI or CT result in this challenge to confirm structural stenosis.

D09-3 | Next step

Discuss appropriate activity and rehabilitation guided by symptoms and function. If a vascular explanation remains possible, pursue appropriate assessment, including ABI when indicated. New cauda equina symptoms or progressive neurological deficits require urgent assessment.

EDITORIAL NOTES / REPLACEMENT WORDING

07 / Confidence & Reasoning - diagnosis and confidence

Exactly one of the six messages below is shown, using the final submitted answer and its confidence rating. Low = 1-2; Moderate = 3; High = 4-5.

C00 | Confidence summary

Your confidence: [Low / Moderate / High] ([rating]/5).

Bracketed fields above are filled from the learner's final confidence selection.

C-CORRECT-LOW | Correct final diagnosis / low confidence

Your final diagnosis matches the intended working diagnosis. Low confidence can be reasonable while important evidence remains incomplete; review which findings distinguish it from the alternatives.

C-CORRECT-MODERATE | Correct final diagnosis / moderate confidence

Your final diagnosis matches the intended working diagnosis. Moderate confidence leaves room to test alternatives and recognize the limits of a clinical pattern.

C-CORRECT-HIGH | Correct final diagnosis / high confidence

Your final diagnosis matches the intended working diagnosis. High confidence should reflect the evidence you established, rather than recognition of a familiar label alone.

C-INCORRECT-LOW | Incorrect final diagnosis / low confidence

Your low confidence signals unresolved uncertainty. Compare the distinguishing findings in the debrief and identify which question would have changed your leading hypothesis.

C-INCORRECT-MODERATE | Incorrect final diagnosis / moderate confidence

Your final diagnosis differs from the intended working diagnosis. Revisit how the pattern of provocation and relief should influence the differential before settling on one explanation.

C-INCORRECT-HIGH | Incorrect final diagnosis / high confidence

Your final diagnosis differs from the intended working diagnosis despite high confidence. Review the competing explanations and ask which findings support or challenge your preferred diagnosis.

EDITORIAL NOTES / REPLACEMENT WORDING

08 / Evidence feedback, submission messages and reading

One evidence message below is appended to the confidence message on page 7. The efficiency paragraph follows in every debrief.

E01 | Key discriminating evidence gathered

Condition: H01 plus either H02 or H03 was investigated.

You investigated both symptom provocation and the effect of posture or sitting. Those are useful discriminators, but they do not replace assessment of vascular causes and red flags or establish imaging confirmation.

E02 | Key discriminating evidence incomplete

Condition: the combination required for E01 was not investigated.

You had not yet established both the trigger pattern and posture-related relief. A correct early guess does not by itself demonstrate well-supported reasoning; these are useful areas to investigate before becoming certain.

E03 | Efficient inquiry - always shown

Efficient inquiry prioritizes findings that separate plausible diagnoses, then tests alternatives and safety concerns. Further questions are useful when they address a specific remaining uncertainty; no points are awarded or deducted for your inquiry or differential revisions.

F01 | Incorrect first attempt

Not quite. One diagnostic attempt remains. You can investigate further and revise your differential before trying again.

F02 | Debrief availability

Your debrief is available below.

F03 | Missing diagnosis or confidence

Enter your final diagnosis before submitting.

Select a confidence rating from 1 to 5 before submitting.

R01 | Clinical reading links shown in the debrief

AAOS: Lumbar spinal stenosis https://www.orthoinfo.org/diseases--conditions/lumbar-spinal-stenosis/

AHA: Diagnosing peripheral artery disease https://www.heart.org/en/health-topics/peripheral-artery-disease/diagnosing-pad